A MORPHISM FROM AN ALGEBRAIC SPACE TO A FORMAL ALGEBRAIC SPACE IS REPRESENTABLE BY ALGEBRAIC SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Base change of a representable surjective etale map of functors is again representable surjective etale. Spaces, Lemma spaces-lemma-base-change-representable-transformations-property, invoked at proof lines 2737-2739. Makes U x_Y Z -> X x_Y Z a representable surjective etale map.
- The diagonal of a formal algebraic space is representable by algebraic spaces and the fibre product with a scheme is a scheme. Same chapter, Lemma lemma-diagonal-formal-algebraic-space, invoked at proof lines 2740-2741. Shows that U x_Y Z is a scheme, so the covering of X x_Y Z is by a scheme.
- The bootstrap theorem producing an algebraic space out of an etale covering by a scheme. Bootstrap, Theorem bootstrap-theorem-final-bootstrap, invoked at proof lines 2742-2743. Upgrades the etale covering by a scheme to the conclusion that X x_Y Z is an algebraic space.
- The notions of formal algebraic space and of a morphism representable by algebraic spaces. Same chapter, Definition definition-formal-algebraic-space (context span, source lines 2634-2649) and the Bootstrap chapter sections on representable morphisms. Fix the meaning of the objects and of the property that is to be proved.

Source reference. Original theorem 04S6: sources/stacks-chow/tags.txt, source lines 6209–6209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIP: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2667–2667. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02WL: sources/stacks-chow/tags.txt, source lines 3808–3808. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism from an algebraic space over S to a formal algebraic space over S. Then f is representable by algebraic spaces.*

Proof. Let $Z \to Y$ be a morphism where Z is a scheme over S. We have to show that $X \times_Y Z$ is an algebraic space. Choose a scheme U and a surjective étale morphism $U \to X$. Then $U \times_Y Z \to X \times_Y Z$ is representable surjective étale (Spaces, Lemma 02WL) and $U \times_Y Z$ is a scheme by Lemma 0AIP. Hence the result by Bootstrap, Theorem 04S6. □